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MILITARY SPECIFICATION

INSIGNIA, EMBROIDERED, ORGANIZATIONAL INSIGNIA AND BADGES, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for Schiffli Embroidered Organizational Insignia, Badges, and Branch of Service Insignia.

1.2 Classification. The insignia covered by this specification shall be of the following types, styles, and classes as specified (see 6.2).

Type I - Organizational Insignia
- Style 1 - Army
- Style 2 - Air Force
- Class 1 - Full Color
- Class 2 - Subdued

Type II - Badges
- Style 1 - Army
- Style 2 - Air Force
  - Class 1 - Ultramarine Base Cloth
  - Class 2 - Olive Green Base Cloth

Type III - Insignia, Branch of Service
- Style 1 - Army
- Style 2 - Air Force
  - Class 1 - Ultramarine Base Cloth
  - Class 2 - Olive Green Base Cloth

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, VA 22304-5050, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8455

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
UU-P-268 - Paper, Kraft, Untreated, Wrapping
CCC-C-461 - Cloth, Uniform Twill, Cotton
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup
PPP-T- 45 - Tape, Gummed, Paper, reinforced and plain, for sealing and securing

STANDARDS

FEDERAL

FED-STD-191 - Textile, Test methods
FED-STD-751 - Stitches, Seams and Stitches

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

THE INSTITUTE OF HERALDRY

A-1 (Series) Army Shoulder Sleeve Insignia, Various
A-13 (Series) Badges, Identification, Various

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B-2 (Series) Insignia Branch of Service, Various
B-5 (Series) Badges, Identification, Various
B-15 (Series) USAF Command/Organizational Patches, Various

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

THE COLOR ASSOCIATION OF THE UNITED STATES

The Standard Color Reference of America, Tenth Edition.
United States Army Color Card, The Official Colors for Branches, Agencies, etc.
DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1974.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia PA 19103.) and (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Guide samples. Unless otherwise specified (see 6.3), samples shall be furnished by the contracting officer to be used for guidance, information and as a standard for colorfastness. Variation from this specification may appear in the guide samples in which event the specification shall govern.

3.2 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Materials. Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Base cloth. The base cloth for all insignia shall be cotton or 50/50 or 65/35 cotton/polyester blend three up, one down, left hand twill, 7.0 to 8.3 ounces per square yard, single ply warp and filling, with a breaking strength of not less than 150 lbs in the warp and 80 lbs in the filling. The residual shrinkage shall be not more than 2 percent in either the warp or filling direction and the cloth shall have not less than fair colorfastness to laundering when tested as specified in 4.5.2.1.

3.3.2 Buckram. The buckram or backing material, where specified, shall be a light, open-weave, single ply fabric weighing 2.5 ounces (minimum) per square yard having a minimum thread count of 40 ends in the warp and 20 ends in the filling. The buckram shall have a minimum breaking strength of 30 pounds in the warp direction and 20 pounds in the filling direction. Testing shall be in accordance with 4.5.2.1.

3.3.3 Embroidery yarn. The embroidery yarn shall be a semidull continuous filament, regenerated cellulose type rayon, two-ply schiffli type yarn. The yarn shall be 150, 100 or 75 denier as specified on the applicable cartoon or drawing. Unless otherwise specified, the use of polyester is authorized in lieu of rayon except when 75 denier is specified. When polyester yarn is used, 135 denier will be substituted for 150 denier, 90 denier will be substituted for 100 denier, and 70 denier will be substituted for 75 denier rayon.

3.3.3.1 Color. The color shall be as specified on the applicable drawing or cartoon. When overedge stitching is required, the color shall be the same as the embroidered border. The use of sulphur dyes is prohibited.

3.3.3.2 Matching. The color shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 plus or minus 500 K, with illumination of 100 plus or minus 20 foot candles, and shall be a good match to the sample under incandescent lamplight at 2300 plus or minus 100 K. The degree of luster shall be equal to the standards established in the DOD Standard Color Card of Official Standardized Shades for Heraldic Yarns for Embroidery and Ribbons, 1974.

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3.3.3.3 Colorfastness. When tested as specified in 4.5.6, the dyed embroidery yarn shall show good fastness to wet-dry cleaning, bleaching, or light, and fair fastness to laundering except that for black, blue, and red shades, the fastness to bleaching shall be not less than fair as defined in FED-STD-191.

3.3.4 Base cloth.

3.3.4.1 Color. The color of the base cloth shall be as follows unless otherwise specified on the applicable drawing or cartoon.

Type	Color of embroidered design edge	Color of base cloth
Type I, styles 1 and 2, class 1	Shades of maroon and red	Khaki
	White	White
	Ultramarine blue	Ultramarine blue
	All other blues and purple	Navy blue
	All shades of black, brown and green	Army green 159
	All shades of yellow, orange, tan and gray or combination of colors.	Khaki
Type I, styles 1 and 2, class 2	All insignia	Olive green 107
Type II and III style 1 and 2, class 1	All insignia	Ultramarine blue
Type II and III, style 1 and 2, class 2	All insignia	Olive green 107

3.3.5 Yarn and thread.

3.3.5.1 Bobbin yarn. The bobbin yarn used in the Schiffli embroidery shall be a continuous multi-filament nylon or two-ply cotton yarn. The yarn shall be undyed except that the yarn for type I, insignia may be dyed black. When the bobbin yarn is dyed black, the yarn shall show fair fastness to wet-dry cleaning, bleaching and laundering as defined by FED-STD-191. The use of sulphur dyes is prohibited. Testing shall be in accordance with 4.5.2.1.

3.3.5.2 Overedge stitching yarn and thread. The yarn and thread used for overedge stitching shall be 600 or 1200 denier multi-filament bright rayon for cover thread (see 3.5.7) and type I, ticket No. 00, 3 ply of V-T-276 for needle and looper threads.

3.3.5.3 Color. The color of the yarn and the thread used for overedge stitching shall be as specified for the embroidered edge on the applicable drawing or cartoon. The use of sulphur dyes is prohibited.

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3.3.5.4 Colorfastness. When tested as specified in 4.5.6, the dyed overedge stitching yarn and thread shall show good fastness to wet-dry cleaning, light, and fair fastness to laundering except that for black, blue, and red shades, the fastness to bleaching shall be not less than fair as defined in *FED-STD-191*.

3.4 Design. The design and color for the background and device of the insignia shall be as shown on the original design drawing or applicable Government cartoon.

3.5 Construction. The insignia shall be embroidered on a Schiffli embroidery machine in accordance with this specification. When furnished by the Government the applicable cartoon shall form a part of this specification. Designation of a specific embroidery span on a Government cartoon does not constitute a mandatory requirement, if all other requirements are met. Cartoons are furnished for the placement of different types of stitches. The cartoon will normally indicate the number of under, cover and miscellaneous stitches to produce an insignia as well as a mandatory requirement for total minimum stitches. The manufacturer, at his discretion, may vary the under and cover stitches provided he meets the total minimum stitches required and produces a product equal to or better than, the standard or guide sample. The density of cover stitches may be increased in critical areas where defects may occur. All thread ends shall be trimmed.

3.5.1 Embroidery stitches. The embroidery stitches shall be of the type, length, number and density specified on the applicable Government cartoon. The total stitches specified on the cartoon are full stitches and constitute two revolutions or movements of the machine. Underlay stitches shown on the cartoon by numbers are 1/2 stitches (one movement of the machine) but are converted to full stitches on the tabular listing. Except where otherwise indicated in the cartoon, all letters shall be underlaid and formed with blatt stitches. When required by contract provisions, the contractor shall submit a sample for approval. When a contractor's sample has been specifically approved for appearance of the embroidery stitching prior to production, deliveries in this respect shall be equal to the approved sample.

3.5.2 Background stitches. Insignia requiring background stitching have either horizontal or vertical background stitching (as indicated on applicable Government cartoon) and the stitch shall be 6/32 inch in length with a plus or minus tolerance of 1/32 inch interlocking in evenly spaced rows (satin fill). Horizontal background stitching shall run in a vertical row and vertical background stitching to run in a horizontal row. The visible width of the embroidered row shall not be more than 5/64 inch. The number of visual rows and the density (number of stitches per inch in a stitch line) shall be as indicated on the applicable Government cartoon.

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3.5.3 Device stitches. Unless otherwise specified, device stitches shall be in accordance with the applicable Government cartoon with respect to spacing the rows, number of visible rows and length and density of stitches. Translation of curvatures and other device features shall be executed with embroidery stitching finished evenly to effect a clean outline.

3.5.4 Underlay stitches. When underlay stitches are required, the number of stitches shall be as indicated on the cartoon. Where underlay is required in areas of design showing rows of stitches, the underlay stitches shall be inserted in a criss-cross pattern. In other areas requiring underlay, the underlay stitches shall be inserted in the direction most suitable for proper design execution.

3.5.5 Stiele stitches. The length and density of required stiele stitches shall be in accordance with the applicable Government cartoon. The stiele stitches shall be tightly drawn and well executed to produce a clear, sharp edging.

3.5.6 Border stitches. All type I insignia shall be bordered by a schrage stitch or a combination of a blatt stitch with a stiele stitch provided it is equal in density to the specified schrage stitch.

3.5.7 Overedge stitching. Unless indicated on the drawing that the insignia is not designed for overedge stitching, an overedge stitching shall be superimposed over the border of all type I, style 1 insignia using yarn and thread specified in 3.3.5. The overedge stitching shall conform to stitch type 504 or 505 of FED-STD-751 except two ends of 600 denier or one end of 1200 denier shall be use for the cover threads. These shall be 24 stitches, plus or minus 4 stitches per inch and the stitch length shall be 1/8 inch, plus or minus 1/32 inch. Tolerance on width shall be acceptable only if the overedge stitching completely covers the border stitching when the specified border on the drawing is 1/8 inch in width. On circular or oval designs, there shall be a half inch overlap of the overedge stitching centered on the bottom of the insignia. The overedge shall start and finish at the bottom point for pointed shields. On round shield or similar designs with the bottom of the insignia round, the overedge shall start and finish in the upper left corner. Stitching for triangles, square, rectangular, polygonal, cylindrical or barrel shape designs shall start and finish on any corner. Where practicable, overedge stitching shall be continuous from start to finish. Unless shown on the drawings, the rounding of corners or angles shall not be permitted. If stitching runs off the edge to begin a new direction for corners, the stitching ends shall be trimmed before being locked in by overedge stitching in the new directions. The final free end of the overedge stitching shall be brought to the back of the insignia and secured with adhesive or fused with a hot iron. The free end shall not be less than 1/2 inch in length and no adhesive shall be permitted within 1/8 inch of the finished insignia edge.

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3.5.7.1 Optional methods of securing final free end. At the option of the contractor, the final free end of the overedge stitching may be secured by blind-stitching 1/4 inch or less of the terminal end to the back of the insignia by hand or machine; or where size and configuration permit the final end may be secured by application of a 3/8 inch by 1 inch (approximate) rectangle of masking tape. No part of the tape shall be within 1/8 inch of the finished edge of the insignia.

3.6 Stiffener. Type I, insignia shall be treated on the reverse side with a fabric finish which will stiffen the insignia to facilitate subsequent processing. The fabric finish shall not affect the colors on the face of the insignia, cause puckering, seriously affecting appearance of insignia, and shall not react adversely to storage, laundering wet-dry cleaning or ironing. The use of buckram specified in 3.3.2 shall be permitted provided that it does not protrude at any point beyond the outline of the finished insignia. A clear plastic type backing fused into the material shall be permitted for use provided it is not heat transferable when pressed at temperatures up to 302 degrees fahrenheit.

3.7 Cutting.

3.7.1 Type I. Type I, insignia shall be cut in single units and the base cloth shall be evenly trimmed as close to the outside of the embroidered edge as is possible without cutting the stitching. The base cloth shall be trimmed close enough to insure the overedge stitching covers the embroidered border when required.

3.7.2 Type II and III. Type II and type III insignia shall be cut so that a minimum of 5/8 inch base material is available between the design and raw edge.

3.8 Tolerances. Unless otherwise specified, the overall measurements of the finished insignia shall be within a tolerance of plus or minus 1/8 inch of the required measurements. Where no tolerances are cited, design dimensions shall have a tolerance of plus or minus 1/32 inch.

3.9 Workmanship. The finished insignia shall be clean and well made and shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or

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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) in the specification.

4.4 First article inspection. Inspection and testing of the first article sample shall be made on a completely finished insignia for all provisions of this specification applicable to the end product examination and tests.

4.5 Quality conformance inspection. Inspection shall be in accordance with provisions set forth in MIL-STD-105, except where otherwise indicated.

4.5.1 Quality assurance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105 except when otherwise specified hereafter.

4.5.2 Component and material inspection. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials as specified herein.

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4.5.2.1 Testing of components. Components and materials shall be tested as specified in table I. The methods of testing specified in FED-STD-191 wherever applicable, and as specified herein shall be followed. The sample size for each characteristic of each component shall be as follows:

Lot size (unit)	Sample size
800 or less	2
801 to 22,000	3
22,001 to over	5

The lot shall be unacceptable if one or more units of any component fail to meet the requirements in table I. The unit of expressing lot size and sample unit shall be as follows:

Component	Lot size expressed as	Sample size for testing
Buckram	Yard	1 linear yard full width
Embroidery yarn	pound	1 holder for every color
Bobbin yarn	pound	5 bobbins for dyed yarn 1 bobbin for natural yarn
Cotton or polyester/cotton base cloth (as applicable)	yard	3-1/2 yards full width

When the data in the "#determ per s/u" and "Results reported as" columns are not specified in table I, they shall be as required in the referenced test method.

TABLE I. Component tests.

Component	Characteristic	Rqm't para	Test method	#determ per s/u	Results reported as
Embroidery yarn	Material ident	3.3.3 1/	1500 or 1520	-	To nearest denier
	Ply	3.3.3	Visual	1	Pass or fail
	Denier	3.3.3	4021	1	To nearest denier
	Presence of Sulphur	3.3.3.1 1/	2020	1	Pass or fail
	Color 2/	3.3.3.1.	Visual	1	Pass or fail
Base cloth	Material ident	3.3.1	1/	-	-
	Weave	3.3.1	Visual	1	Pass or fail
	Weight	3.3.1	5041	5	Nearest 0.1 oz/sq
	Ply	3.3.1	Visual	1	Pass or fail
	Breaking strength:				
	Warp (lbs)	3.3.1	5100	5	Nearest 1 lb.
	Filling (lbs)	3.3.1	5100	5	Nearest 1 lb.

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TABLE II. Component tests.

Component	Characteristic	Rqm't para	Test method	#determ per s/u	Results reported as
	Shrinkage:				
	Warp	3.3.1	5550	3	Nearest 0.1%
	Filling	3.3.1	5550	3	Nearest 0.1%
	Colorfastness to Laundering	3.3.1	5610	1	Pass or fail
Bobbin yarn	Material ident	3.3.5 1/	1200 & 1530	1 -	Pass or fail -
	Ply (when cotton yarn is used)	3.3.5	Visual	1	Pass or fail
	Color (when applicable) 2/	3.3.5	Visual	1	Pass or fail
	Presence of sulphur	3.3.5 1/	2020	1	Pass or fail
	Colorfastness to (when applicable):				
	wet-dry cleaning	3.3.5	5622	-	-
	Bleaching	3.3.5	5600	-	-
	Laundering	3.3.5	5614	-	-
Buckram	Weave	3.3.2	Visual	1	Pass or fail
	Weight	3.3.2	5041	5	Nearest 0.1 oz/sq yd
	Yarn count:				
	Warp	3.3.2	5050	5	Nearest whole number
	Filling	3.3.2	5050	5	Nearest whole number
	Breaking strength:				
	Warp (lbs)	3.3.2	5100	5	Nearest 1 lb.
	Filling (lbs)	3.3.2	5100	5	Nearest 1 lb.
Overedge stitching yarn	Material ident	3.3.5	1500 or 1520	-	-
	Ply	3.3.5	Visual	1	Pass or fail
	Denier	3.3.5	4021	1	To nearest denier
	Presence of Sulphur	3.3.5 1/	2020	1	Pass or fail
	Color	3.3.5	Visual	1	Pass or fail

1/ Unless otherwise specified a contractor's certification of compliance may be accepted as evidence of conformance.

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2/ When Government certified yarn is used, this test may be omitted by contractor (see 6.6). The Government reserves the right to draw samples for verification.

4.5.3 Visual examination of the end item. Defects found during this examination shall be assigned from 1 to 3 demerit points, as prescribed in table III, and all demerit points found shall be added accordingly. All defects shall be so scored, except where two or more classifications of table III are applicable for a single location condition, in which case the highest point count applicable shall be scored. The lot size, sample size and sample unit shall be expressed in units of one finished insignia. The sample size and maximum sum of points permitted shall be as shown in table II, expressed as maximum sum of points for 3 and 2 point defects combined, and for all defects combined. The lot shall be unacceptable if the sum of the point value of the examined sample exceeds the limits established in table II. Normal inspection distance for workmanship shall be at arm's length (16 to 22 inches). Examination for other defects may be determined at an inspection distance suitable for such examinations.

TABLE II. Maximum sums of points

Lot size	Sample	3 & 2 points combined	3, 2 and 1 points combined
(Regular inspection)			
Under 3200	125	25	35
3201 - 10,000	200	35	53
10,001 - 35,000	315	53	75
35,001 - 150,000	500	75	110
150,001 and up	800	110	150
(Tightened inspection)			
3201 - 10,000	200	30	45
10,001 - 35,000	315	45	68
35,001 - 150,000	500	68	103
150,001 and up	800	103	138

TABLE III. Defects

Examine	Defect	Point count
Base cloth	Other than specified	3 - -
	Any splice, cut, hole or tear visible on face.	3 - -
	Color shade not as specified	3 - -

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TABLE III. Defects - continued.

Examine	Defect	Point count		
Stiffener/backing	Omitted or incomplete when required.	3	-	-
	Back not coated with rubber base adhesive when required.	3	-	-
	Coating not uniform or incomplete	-	-	1
	Buckram protrudes more than 1/32 inch beyond finished insignia.	-	-	1
	Stiffener visible on front.	3	-	-
	Stiffener/backing not as specified.	3	-	-
	Design in adequately covered with base cloth, background, or underlay showing through (thin or uneven stitch density).	3	-	-
Construction (embroidery)	Embroidery yarn not an acceptable match . . .	3	-	-
General	Background not as specified	3	-	-
	Device not as specified	3	-	-
	Background not interlocked in evenly spaced rows	-	-	1
	Spacing of rows not in accordance with cartoon	-	2	-
	Non-uniform embroidery stitch density	-	-	1
	Bobbin thread showing on face	-	2	-
	Tight tension-distorted surface of face side.	3	-	-
	Loose tension-loopy surface	-	*	-
	Traveling (walking) stitches on face of insignia	-	*	-
	Broken stitches or stitching incomplete (missing).	-	*	-
Design	Any departure from cartoon or drawing	3	-	-
	Design or device poorly defined, sharp outline lacking.	3	-	-
	Shift, device or outline misshaped misplaced or overlapped incorrectly.	-	*	-
	Device flat-underlay missing or not well rounded or prominent, underlay insufficient.	-	-	1
	Border not constructed of schrage stitches or a combination of blatt and stiele stitches when required.	3	-	-
	Curvature or design features fuzzy, not clearly outlined	-	2	-
	Stiele stitches not firmly drawn to produce clearly defined rows or edges.	-	2	-
	Stiele stitch in arcs, bars or device not as required by cartoon.	-	2	-

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TABLE III. Defects - continued.

Examine	Defect	Point count		
Workmanship	Kinked, bunched, tangled yarn on face	-	*	-
General	Broken, scuffed, or cut yarns on face	-	*	-
	Spot, stain, stiffener, or foreign matter on face	-	-	1
	Two or more thread ends untrimmed	-	-	1
	More than two stumps longer than 1/16 inch. .	-	-	1
	Creased or folded, will not flatten by manual pressure	-	*	-
	Not paired, left and right, when required . .	3	-	-
	Embroidery yarn not an acceptable match when paired.	3	-	-
Finish	Omitted completely or not as specified. . . .	3	-	-
(border and edges)	Final free end not secured on back of insignia			
(Type I)	as specified	-	2	-
	Final free end not specified length	-	-	1
	Overedge does not adequately cover border . .	-	2	-
	Bunched or tangled yarns.	-	2	-
	Broken or skipped stitches.	-	2	-
	Raveled, ends not secured	-	2	-
	Run-off, not caught in base cloth	-	2	-
	Start and finish not at specified point . . .	-	-	1
	Angles or pointed corners rounded off	-	-	1
	Tight or loose tension.	-	-	1
	Overedge less than 3/32 or more than 5/32 inch length	-	-	1
	Continuity lacking when design required continuity	-	-	1
	Adhesive from final end exposed on finished edge	-	-	1
	Cut schrage or border stitches not adequately covered by overedge stitching.	3	-	-
	Not finished as specified	3	-	-
Borders	Not uniform in width or unevenly trimmed. . .	-	2	-
(base cloth)	Corners or points not well defined.	-	2	-
Types II and III	Free material between edge design and raw edge of material not as specified.	3	-	-
	Not cut in single units when required	3	-	-

*Defects designated by an (*) asterisk shall be classed as follows:
3 point defect when seriously affecting appearance or serviceability.
2 point defect when affecting appearance but not seriously.
1 point defect when affecting serviceability but not seriously.

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4.5.4 Examination for dimensions. Measurements shall be taken of all points, lines, areas, curves and spaces for which dimensions are specified in the drawing, including:

Overall length	Overall width
Border width	Base material around insignia
Device location	Device dimension
Spaces and distances between intermediate points	

The lot size shall be expressed in units of one insignia each for individual insignia. The sample unit shall be one completely fabricated insignia. Each measurement found to be outside of the specified tolerances shall be scored as a defect. The total defects found shall not exceed the maximum of the table in this paragraph. The sample shall be selected at random from among the units selected for visual examination. The tolerances for paragraph 3.8 or any other specific tolerances provided shall be applied.

Lot size	Sample size	No. of defects (Max)
Under 3200	5	0
3201 to 22,000	15	1
over 22,000	25	2

4.5.5 Inspection of packaging. An examination shall be made to determine that the packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table IV. The sample unit shall be one shipping container fully prepared for delivery with the exception it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the Acceptable Quality Level (AQL) shall be 4.0 defects for per hundred units.

TABLE IV. Classification of preparation for delivery defects.

Examination	Defects
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as, incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling.
	Bulging or distortion of containers.

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TABLE IV. Classification of preparation for delivery defects-continued.

Examination	Defects
Weight	Net weight exceeds requirements.
Preservation (when applicable)	Omitted or not uniformly distributed in container.

4.5.5.1 Examination of count. An examination shall be made to determine that the number of insignia per bundle, package, set-up box and shipping container is as required by section 5 of this specification. Table V shall govern this examination.

TABLE V. Examination of count.

Examine	Unit of product or lot expressed in units of	Sample size	
		#Units in lot	#Units selected
Unit Packaging			
a. Number of insignia per bundle	One bundle	Up to 300 301 - 500 501 - 800 801 - 1300 1301 & over	5 10 15 25 30
Intermediate packaging.			
b. Number of bundles per set up box	One set up box	Up to 300 301 - 500 501 - 800 801 - 1300 1301 & over	5 10 15 25 30
Packing.			
c. Number of set -up boxes per shipping container	One shipping container	Up to 3 4 to 6 7 & over	All 3 4

The lot shall be unacceptable if any of the defects listed below are found:

a. Number of insignia per bundle. The average number of insignia per bundle is less than specified

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b. Number of bundles per set-up box. The number of bundles in any set-up box is less than specified.

c. Number of set-up boxes per shipping container. The total number of insignia in a shipping container is less than the total indicated on outside container marking.

d. Number of insignia per shipping container. The number of insignia marked on the shipping container is more than the total of insignia marked on the packages in the container.

4.5.6 Testing of the end item. The end item shall be tested as specified in table VI. The lot size shall be expressed in units of one insignia. The unit of product shall be eight (8) insignia. The lot shall be unacceptable if one or more units fail to meet the requirements referenced in table VI. The sample size shall be as follows:

Lot size (each)	Sample size
22,000 or less	2
22,001 and over	3

TABLE VI. End item tests.

Characteristic	Requirement paragraph	Test method
Embroidery and overedge stitching yarn:		
Colorfastness to:		
Wet-dry cleaning	3.3.3.3 and 3.3.5.4	5622
Bleaching	3.3.3.3 and 3.3.5.4	5600
Light	3.3.3.3 and 3.3.5.4	5660 1/
Laundering 2/	3.3.3.3 and 3.3.5.4	5610
Overedge stitching thread:		
Stitches per inch	3.5.7	5052 3/
Fabric finish:		
Plastic backing	3.6	4/

1/ In the event of dispute resulting from the test with method 5660 or a result of anomalous behavior of certain dye types or formulations, the contracting officers may authorize the exposure to natural light in accordance with test method 5662.

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2/ Only the stain on the fiber of the color transfer cloth represented in the basic materials being tested shall be evaluated.

3/ The contractor shall furnish a certificate of compliance.

4/ Test will be accomplished by ironing the insignia against cloth material at required temperature. If plastic transfers to the cloth, the entire lot shall be unacceptable.

4.5.7 Palletization examination. An examination shall be made to determine that the palletization is in compliance with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Insignia of one type, style, and class shall be put up in bundles of twenty each and securely tied, or cross tied, with cotton twine or thread. Unless otherwise specified, insignia of one type, style, and class may be placed in a clear polyethylene bag or blister pack. Closure of each bag shall be by heat-sealing or tape closure.

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5.1.1.2 Intermediate packaging. Ten bundles (200 insignia) of one type, class and style preserved as specified in 5.1 shall be placed in a set-up paperboard box conforming to type I, variety 1, class A, style 3 or 4 of PPP-B-676. Closure shall be in accordance with the appendix of the container specification.

5.1.2 Level C. Insignia shall be preserved to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. Insignia shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B, C or commercial as specified (see 6.2).

5.2.1 Level A. Insignia preserved, as specified, shall be packed in a snug-fitting fiberboard shipping container conforming to class weather-resistant, V3a or V3c of PPP-B-636. Closure shall be in conformance with the appendix of the appropriate container specification. The weight of the contents of each shipping container shall not exceed the size and weight limitation of the container specification.

5.2.2 Level B. Insignia, preserved as specified in 5.1 shall be packed in a snug-fitting fiberboard box conforming to class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with the container specification. The weight in each shipping container shall not exceed the weight and size limitation of the container specification.

5.2.3 Level C. Insignia of one type, style, and class only, preserved as specified in 5.1, shall be packed to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. Insignia shall be preserved in accordance with the applicable requirements of ASTM-D-3951.

5.3 Palletization. Items packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type I, class 1, style 1, size A, wood group I, II, III, or IV of MIL-P-15011, or 4-way, type IV, V or VIII, class 1, style A, size 2, wood group I, II, III or IV, grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate boxes, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM-D-3951, as applicable.

5.4.1 Unit packaging. Unit packaging shall have the required information legibly printed or stamped in black on the bag across the center face, or on a white paper label inserted with in the bag so as to permit ready identification. When unit packaging is secured by cotton twine or thread, the label shall be non-adhesive and secured under the tie. The bag or label shall indicated the following:

Stock Number
Nomenclature
Quantity

6. NOTES.

6.1 Intended use. The insignia covered by this specification are intended to be worn by Department of Defense personnel to designate organization or skills of the wearer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, style, class (see 1.2).
c. Drawing number (see 2.1.2).
d. When a guide sample will not be furnished (see 3.1).
e. When a first article is not required (see 3.2).
f. Cartoon (see 3.5).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
h. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.4). The first article should consist of one completed insignia. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products,

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who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Supersession data. This specification includes, in part, the requirements of MIL-I-3037G dated 30 September 1981 and Amendment 3 thereto dated 31 December 1982.

6.6 Certification of embroidery yarn. Converters and suppliers of embroidery yarns may avail themselves of Government certification of their product with respect to shade matching, luster and colorfastness properties, once it is delivered into Government contractor's plant. Embroidery yarn which has not been so certified may cause delays in the manufacturing of insignia pending completion of Government certification tests. Inquiries about the certification procedures may be addressed to Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101-8419.

6.7 Cross-reference. Previous classifications listed in MIL-I-3037 have been changed to the new classifications indicated in 1.2 this specification. New items may be binned with or substituted for previous items.

Previous classification under MIL-I-3037	New classification under this specification
Material A	
Type I,	Type I
Type VI,	Type II
Not classified	Type III

Custodians:
Army - IH
Air Force - 11
Navy - NU

Review activities:
Air Force - 82,99
Navy - MC

User activities:
Air Force - 45

Preparing activity:
Army - IH

(Project No. 8455 - 0200)

FSC 8455

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-14652	Insignia, Embroidered, Organizational Insignia and Badges, General Specification for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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